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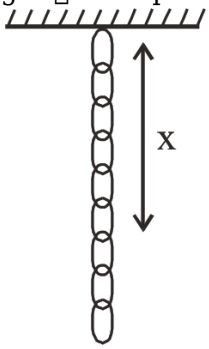
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PHYSICS

1) A uniform chain of mass m and length is suspended from a fixed support. Tension in the chain



at a distance x from the fixed end is :

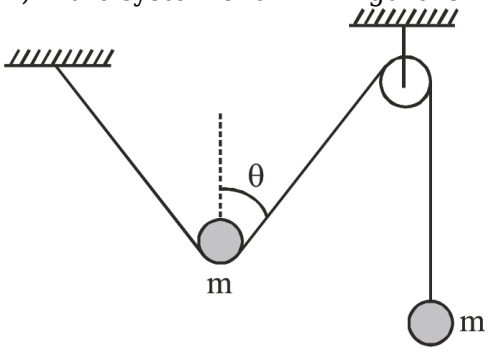
(1) $\frac{m}{\ell}xg$

(2) $\frac{m}{\ell}(\ell - x)g$

(3) $\frac{m\ell}{x}g$

(4) $\frac{m\ell}{(\ell - x)}g$

2) If the system shown in figure is in equilibrium, the value of angle θ is :



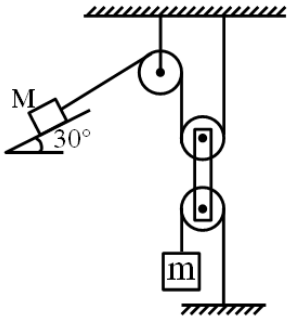
(1) 30°

(2) 60°

(3) 45°

(4) 37°

3) In the arrangement shown, neglect the mass of the ropes and pulley. What must be the value of m



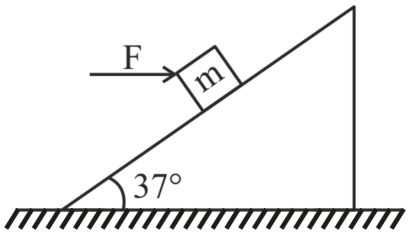
to keep the system in equilibrium ? There is no friction anywhere.

(1) 1

(2) 2 M

(3) $\frac{M}{2}$

(4) $\frac{M}{4}$



4)

A block of mass m is placed on smooth fixed inclined plane.
A force F is applied horizontally to keep block at rest. Find value of F.

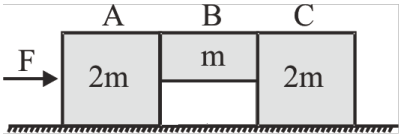
(1) $\frac{mg}{4}$

(2) $\frac{3mg}{4}$

(3) $\frac{mg}{5}$

(4) $\frac{2mg}{5}$

5) The system is pushed by a force F as shown in figure. All surfaces are smooth except between B and C. Friction coefficient between B and C is μ. Minimum value of F to prevent block B from downward slipping is :-



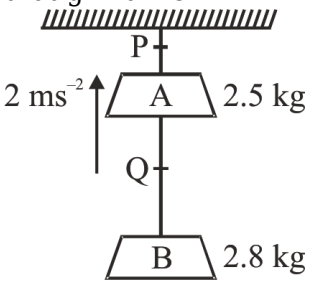
(1) $\left(\frac{3}{2\mu}\right)mg$

(2) $\left(\frac{5}{2\mu}\right)mg$

(3) $\left(\frac{5}{2}\right)\mu mg$

(4) $\left(\frac{3}{2}\right)\mu mg$

6) Two bodies A and B of masses 2.5 kg and 2.8 kg is suspended from a rigid support by two inextensible wires each of length 1.8 m. The lower wire is of mass 1.5 kgm^{-1}, while the upper wire is of negligible mass. If the whole system moves in upward direction with an acceleration of 2 ms^{-2}. Find the tension at the mid-point P of the upper wire and at the mid-point Q of the lower wire, given that g=10 ms^{-2}

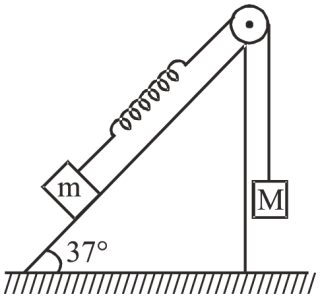


(1) T_P = 96 N, T_Q = 49.8 N

(2) T_P = 96.8 N, T_Q = 49 N

(3) T_P = 24 N, T_Q = 48 N

(4) T_P = 96 N, T_Q = 24 N



7) A block of mass m is attached with a massless spring of force constant K. The block is placed over a fixed rough inclined surface for which the coefficient of friction is $\mu = \frac{3}{4}$. The block of mass m is initially at rest. The block of mass M is released from rest with spring in unstretched state. The minimum value of M required to move the block up the plane is (Neglect mass of string and pulley and friction in pulley) :

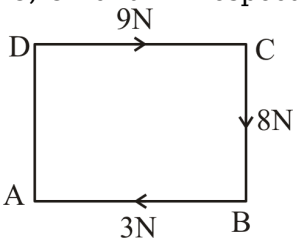
(1) $\frac{3}{5}$ m

(2) $\frac{4}{5}$ m

(3) $\frac{6}{5}$ m

(4) $\frac{3}{2}$ m

8) ABCD is a rectangle. forces of 9N, 8N, 3N act along the lines DC, CB and BA respectively, in the directions indicated by in the figure. Then the resultant force is -

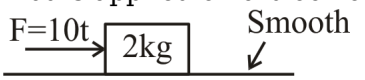


(1) 8N

(2) 5N

(3) 20N

(4) 10N

9) A force F = 10t is applied on a block of mass 2 kg as shown. Distance covered by block in 5 sec. (t is in seconds) 

(1) 104 m

(2) 100 m

(3) 110 m

(4) 120 m

10) Choose the **incorrect** statement :

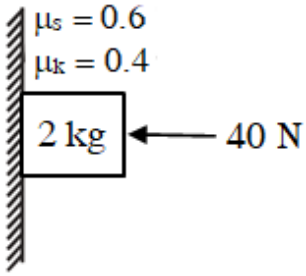
(1) Static friction is self adjusting.

(2) Newton's second law of motion is valid in inertial frame of reference.

(3) Coefficient of static friction is never smaller than coefficient of kinetic friction.

(4) Normal reaction on a man standing in a lift will be more than its actual weight if the lift is accelerating downwards.

11) When block is in contact with wall, friction between wall & block will be. (g = 10 m/s^2) :-



(1) 24 N

(2) 16 N

(3) 20 N

(4) None

12) A box of mass 8kg is placed on a rough inclined plane of inclination θ. Its downward motion can be prevented by applying an upward pull F and it can be made to slide upwards by applying a force 2F. The coefficient of friction between the box and the inclined plane is :-

(1) $\frac{1}{3}\tan\theta$

(2) 3 tan θ

(3) $\frac{1}{2}\tan\theta$

(4) 2 tan θ

13) If the coefficient of friction between an insect and bowl surface is μ and the diameter of bowl is

r. Find the maximum height to which the insect can crawl in the bowl :-

(1) $\left[1-\frac{1}{\sqrt{1-\mu^2}}\right]\times r$

(2) $\left[1-\frac{1}{\sqrt{1+\mu^2}}\right]\times r$

(3) $\left[1-\frac{r}{\sqrt{1+\mu^2}}\right]$

(4) None of these

14) **Assertion :** Magnitude of the contact force is always greater than the magnitude of frictional force.
Reason : Contact force is the resultant of the friction force and normal reaction.

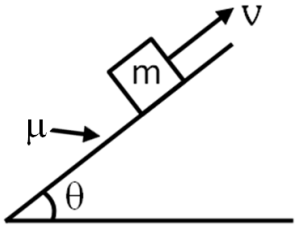
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

15) A block of mass m is thrown upwards with some initial velocity as shown. On the block



Column-I		Column-II	
(a)	Net force in horizontal direction	(P)	zero
(b)	Net force in vertical direction	(Q)	$m(g\sin\theta+\mu g\cos\theta)$
(c)	Net force along the plane	(R)	$m(g\sin\theta\cos\theta+\mu g\cos^2\theta)$
(d)	Net force perpendicular to plane	(S)	$m(g\sin^2\theta+\mu g\sin\theta\cos\theta)$

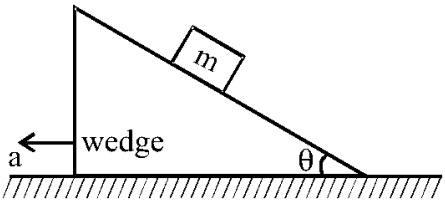
(1) (a)-P, (b)-R, (c)-Q, (d)-P

(2) (a)-R, (b)-S, (c)-Q, (d)-P

(3) (a)-P, (b)-S, (c)-Q, (d)-P

(4) (a)-Q, (b)-R, (c)-P, (d)-R

16) Find the horizontal acceleration of the inclined wedge so that the block placed on the wedge falls freely :-

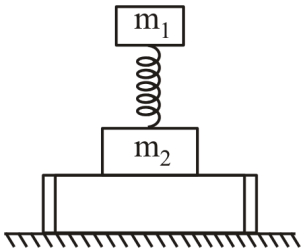


(1) g tanθ

(2) g/tan θ

(3) g cos θ

(4) g sec θ

17) 

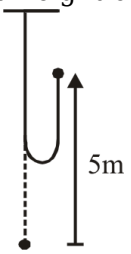
Two block tied by spring are placed on a table.
If table is suddenly removed, find acceleration of m_1 and m_2 just after it -

(1) $0, \frac{(m_1 + m_2)g}{m_2}$

(2) 0, 0

(3) g, g

(4) $\frac{(m_1 + m_2)g}{m_1}, 0$

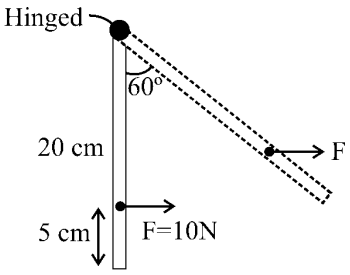
18) A block of mass 2 kg is suspended from a string, is released from a height of 5m as shown in figure then what will be the impulse when string just becomes tight :- 

(1) 2 N-s

(2) 0.2 N-s

(3) 20 N-s

(4) 200 N-s

19) 

On a uniform rod of mass 100 g a constant force $\vec{F}$ of 10 N is acting as shown in the figure. Then find the work done by force $\vec{F}$ in displacing it by an angle 60º :-

(1) $1.5\sqrt{3}$J

(2) $0.75 \times \sqrt{3}$J

(3) 0.75 J

(4) None of these

20) A body of mass 3 kg is under a force, which causes a displacement in it given by S = $\frac{t^3}{3}$(in m). Find the work done by the force in first 2 seconds

(1) 2 J
(2) 3.8 J
(3) 5.2 J
(4) 24 J

21)

A particle of mass m at rest is acted upon by a force P for a time t. Its kinetic energy after an interval t is :

(1) $\frac{P^2t^2}{m}$
(2) $\frac{P^2t^2}{2m}$
(3) $\frac{P^2t^2}{3m}$
(4) $\frac{Pt}{2m}$

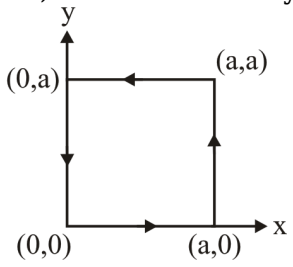
22) A spring of force constant k is cut in two parts at its one third length. When both the parts are stretched by same amount, the work done in the two parts, will be

(1) equal in both
(2) greater for the longer part
(3) greater for the shorter part
(4) data insufficient

23) A particle of mass 2 kg is moving on a rough horizontal surface. If its velocity changes from 20 m/sec to 10 m/sec over a displacement of 30 m, then coefficient of friction for the surface is :-

(1) 0.5
(2) 0.2
(3) 0.4
(4) 0.3

24) The work done by the force $\vec{F} = x^2\hat{i} + y^2\hat{j}$ around the closed path shown in fig. is :-



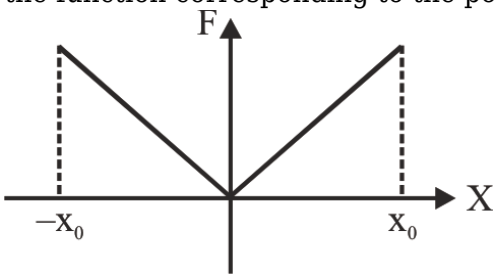
(1) $\frac{2}{3}a^3$

(2) Zero

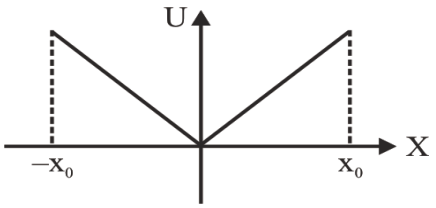
(3) a^3

(4) $\frac{4}{3}a^3$

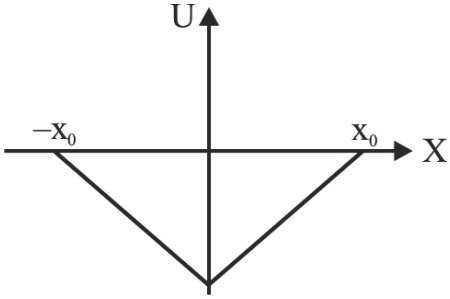
25) Figure shows a plot of the conservative force F in a unidimensional field. The plot representing the function corresponding to the potential energy (U) in the field is



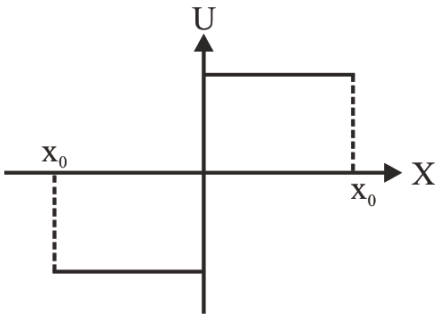
(1)



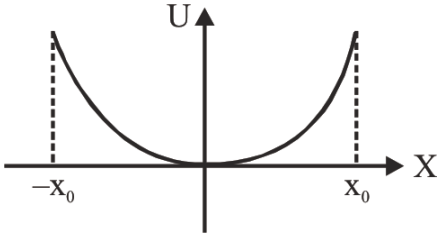
(2)



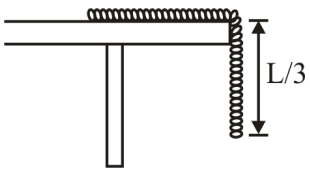
(3)



(4)



26) A uniform chain (mass M, length L) is released from rest from a smooth horizontal surface as shown in the figure. Velocity of the chain at the instant it completely comes out of the table will be :-



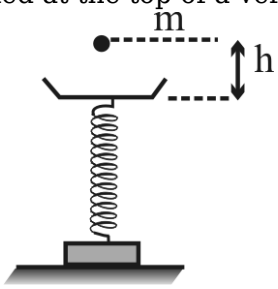
(1) $v = \sqrt{gL}$

(2) $v = \sqrt{\frac{3}{4}gL}$

(3) $v = \sqrt{\frac{8}{9}gL}$

(4) $v = \sqrt{\frac{7}{8}gL}$

27) A ball of mass m is dropped from a height h on a platform fixed at the top of a vertical spring.



The platform is displaced by a distance x. The spring constant is

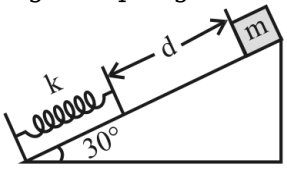
(1) $\frac{2mg}{x}$

(2) $\frac{2mgh}{x^2}$

(3) $\frac{2mg(h+x)}{x^2}$

(4) $\frac{2mg(h+x)}{h^2}$

28) A block of 4 kg mass starts at rest and slides a distance d down a frictionless incline (angle 30°) where it runs into a spring of negligible mass. The block slides an additional 25 cm before it is brought to rest momentarily by compressing the spring. The force constant of the spring is 400 Nm^{-1}. The value of d is (take g = 10 ms^{-2})



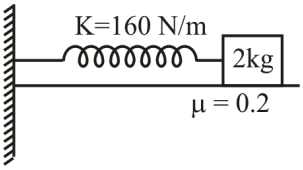
(1) 25 cm

(2) 37.5 cm

(3) 62.5 cm

(4) None of these

29) Find the initial compression in spring so that the block travels a distance of 8m before coming



to instantaneous rest after block is released.

(1) 1.01 m

(2) 2.02 m

(3) 3.03 m

(4) 4.025 m

30) A pendulum of mass 1kg and length = 1m is released from rest at angle θ = 60°. The power delivered by all the forces acting on the bob at angle θ = 30° will be (g = 10 m/s^2) :-

(1) 13.4 W

(2) 20.4 W

(3) 24.6 W

(4) zero

31) Two particles of mass 2kg and 1kg are placed at points (1, –2) and (–4, 3). Where should a third particle of mass 5kg be placed so that centre of mass of three particle system lies at origin ?

(1) $\left(\frac{1}{5}, \frac{2}{5}\right)$

(2) $\left(\frac{2}{5}, \frac{1}{5}\right)$

(3) $\left(\frac{3}{5}, \frac{4}{5}\right)$

(4) $\left(\frac{-2}{5}, \frac{-1}{5}\right)$

32) A circular disc of radius R is removed from a bigger circular disc of radius 2R such that the circumference of the disc coincide. The center of mass of the new disc is αR from the center of the bigger disc. The value of α is :-

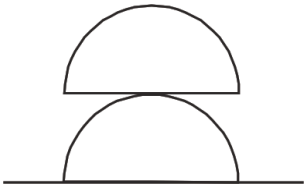
(1) $\frac{1}{3}$

(2) $\frac{1}{2}$

(3) $\frac{1}{6}$

(4) $\frac{1}{4}$

33) A solid sphere of mass m & radius R is cut into two equal parts and kept as shown. Find the

height of centre of mass from the ground. 

(1) $\frac{R}{2}$

(2) $\frac{3R}{8}$

(3) $\frac{7R}{8}$

(4) $\frac{9R}{8}$

34) With respect to the centre of mass of a two body system. :-

(1) The two bodies may move in same direction

(2) The heavier body may move faster

(3) One body may be at rest while the second body is moving

(4) None of these

35) Two cars start at the same point, but travel in opposite directions on a circular path of radius R, each at speed v. While each car travels a distance less than $\frac{\pi}{2}R$ (i.e. less than one quarter of circle), the center of mass of the two cars :-

(1) remains at the initial point.

(2) travels along a diameter of the circle at speed $v' < v$

(3) travels along a diameter of the circle at speed $v' = v$

(4) travels along a diameter of the circle at speed $v' > v$

36) A vessel at rest explodes into three pieces. Two pieces having equal masses fly off perpendicular to one another with the same velocity 30 m/s. The third piece has three times mass of each of other piece. The magnitude and direction of the velocity of the third piece will be :-

(1) $10\sqrt{2}ms^{-1}$ and 135° from either

(2) $10\sqrt{2}ms^{-1}$ and 45° from either

(3) $\frac{10}{\sqrt{2}}ms^{-1}$ and 135° from either

(4) $\frac{10}{\sqrt{2}}ms^{-1}$ and 45° from either

37) Two billiard balls undergo a head-on collision. Ball 1 is twice as heavy as ball 2. Initially, ball 1 moves with a speed v towards ball 2 which is at rest. Immediately after the collision, ball 1 travels at a speed of v/3 in the same direction. What type of collision has occured?

(1) inelastic

(2) elastic

(3) completely inelastic

(4) Cannot be determined from the information given

38) A body of mass m moving with velocity v makes a head on perfectly inelastic collision with another body of mass 2m which is initially at rest. Then percentage loss in the kinetic energy :-

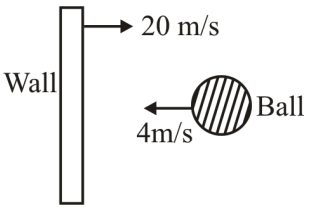
(1) 50%

(2) $\frac{200}{3}\%$

(3) $\frac{800}{3}\%$

(4) 25%

39) Find velocity of ball after collision, if $e = \frac{1}{2}$?



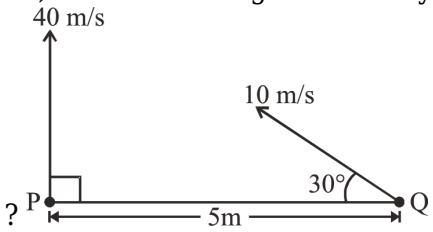
(1) 32 m/s

(2) 16 m/s

(3) 8 m/s

(4) 4 m/s

40) What is the angular velocity of particle P w.r.t. the particle Q in the arrangement shown in figure



(1) 3 rad/s

(2) 7 rad/s

(3) 10 rad/s

(4) 15 rad/s

41) A particle is moving on a circular path of radius $\frac{100}{\sqrt{19}}$ m in such a way that magnitude of its velocity varies with time as $v = 2t^2 + t$, where v is velocity in m/s and t is time in s. The acceleration of the particle at t = 2 s is

(1) 21 m/s^2

(2) 9 m/s^2

(3) 10 m/s^2

(4) 13.5 m/s^2

42) A coin is placed on the horizontal surface of a rotating disc. The distance of the coin from the axis is 1 m and coefficient of friction is 0.5. If the disc starts from rest and is given an angular acceleration $\frac{1}{\sqrt{2}}$ rad / sec^2, then the number of revolutions through which the disc turns before the coin slips is

(1) $\frac{4}{\pi}$

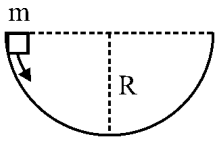
(2) $\frac{7}{4\pi}$

(3) $\frac{5}{4\pi}$

(4) $\frac{3}{4\pi}$

43)

A mass m slips along the wall of a semispherical surface of radius R from rest. Its velocity at the bottom of the surface is :-



(1) $\sqrt{Rg}$

(2) $\sqrt{2Rg}$

(3) $2\sqrt{\pi Rg}$

(4) $\sqrt{\pi Rg}$

44) A stone is rotated in a vertical circle, speed at bottom most point is $\sqrt{8gR}$ where R is the radius of circle. The ratio of tension at the top and the bottom is :-

(1) 1 : 2

(2) 1 : 3

(3) 2 : 3

(4) 1 : 4

45) A tube of length L is filled completely with an incompressible liquid of mass M and closed at both the ends. The tube is then rotated in a horizontal plane about one of its ends with a uniform angular velocity ω. The force exerted by the liquid at the other end is:

(1) $\frac{ML\omega^2}{2}$

(2) $ML\omega^2$

(3) $\frac{ML\omega^2}{4}$

(4) $\frac{ML^2\omega^2}{2}$

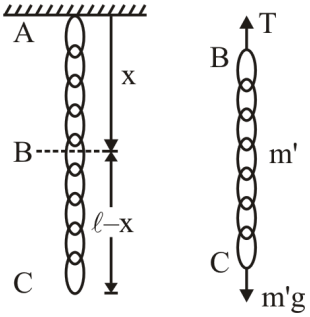
ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**2**	**2**	**3**	**2**	**2**	**1**	**1**	**4**	**1**	**4**	**3**	**1**	**2**	**1**	**2**	**2**	**1**	**3**	**2**	**4**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**2**	**3**	**1**	**2**	**4**	**3**	**3**	**2**	**4**	**1**	**2**	**1**	**3**	**4**	**2**	**1**	**2**	**2**	**1**	**2**
Q.	41	42	43	44	45															
A.	**3**	**2**	**2**	**2**	**1**															

SOLUTIONS

PHYSICS



1)

□ → m

$1 \rightarrow \frac{m}{\ell}$

$(□ - x) \rightarrow \frac{m}{\ell} \times (□ - x) = m'$

T = m'g

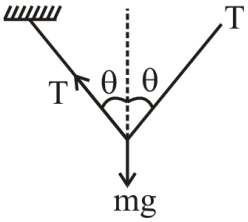
$T = \frac{m}{\ell}(□ - x)g$

2) T = mg ...(1)

2T cosθ = mg ...(2)

by equation (1) and (2) $\cos\theta = \frac{1}{2}$

θ = 60°



3)

Mg sin 30 = T = mg

$m = \frac{M}{2}$

4) F cos 37 = mg sin 37

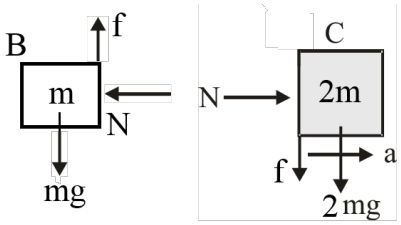
$F\frac{4}{5} = mg\frac{3}{5}$

$\boxed{F = \frac{3mg}{4}}$

5)

Horizontal acceleration of the system is,

$$a = \frac{F}{2m + m + 2m} = \frac{F}{5m}$$



Let N be the normal reaction between B and C.

Free body diagram of C gives

$$N = 2ma = \frac{2}{5}F$$

Now, B will not slide downward if

$\mu N \geqslant m_B g$

or $$\mu\left(\frac{2}{5}F\right) \geqslant mg$$

or $$F \geqslant \frac{5}{2\mu}mg$$

So, $$F_{min} = \left(\frac{5}{2\mu}\right)mg$$

6)

mass of rope, m = 1.5 × 1.8 = 2.7 kg

T_P = (2.5 + 2.7 + 2.8) (g + a) = 96 N

T_Q = (2.8 + $\frac{2.7}{2}$) (g + a) = 4.15 × 12 = 49.8 N

7)

Maximum extension in spring x = $\frac{2Mg}{K}$

⇒ Kx = mg sinθ + μ mgcosθ

⇒ $2Mg = mg \times \frac{3}{5} + \frac{3}{4}mg \times \frac{4}{5}$

⇒ $M = \frac{3}{5}m$

8) $\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$

$= 9\hat{i} - 8\hat{j} - 3\hat{i} = 6\hat{i} - 8\hat{j}$

$|\vec{F}_{net}| = \sqrt{6^2 + 8^2} = 10$

9) $a = \frac{10t}{2} = 5t$

$$\int_{v=0}^{v} dv = \int_{t=0}^{t} 5t.dt$$

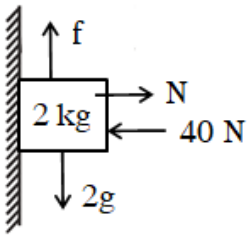
$$V = \frac{5t^2}{2}$$

$$\int_{x_i}^{x_f} dx = \int_{t=0}^{t=5} \frac{5t^2}{2}, dt$$

$$x_f - x_i = \left[\frac{5}{2} \cdot \frac{t^3}{3}\right]_0^5$$

$$S = \frac{625}{6} \simeq 104.1$$

10) Option (4) is incorrect N = m(g – a)



11)

$f_{□}$ = μ_sN = 0.6 × 40 = 24 N

2g = 20 N

as 20 < $f_{□}$

thus f = 20 N

12) During downward motion :

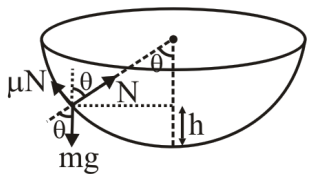
F = mg sin θ – μmg cos θ

During upward motion :

2F = mg sin θ + μmg cos θ

Solving above two equations, we get;

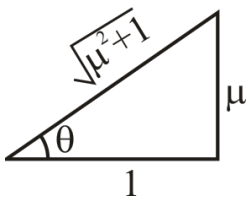
$$\mu = \frac{1}{3}\tan\theta$$



13)

N = mg cos θ & mg sin θ = μN

⇒ sin θ = μ cos θ ⇒ tan θ =μ



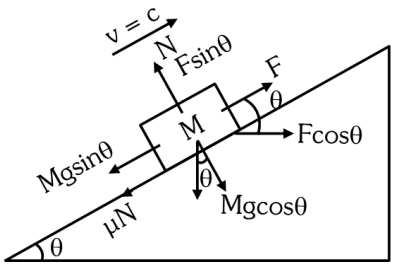
h = r (1– cos θ) = r $\left(1 - \frac{1}{\sqrt{1+\mu^2}}\right)$

14) $F_C = \sqrt{f^2 + N^2}$

Contact Force And Angle Of Friction

$F_C = \sqrt{f^2 + N^2}$

15) F = Mgsinθ + μN = Mgsinθ + μMgcosθ



Net force in horizontal direction = Fcosθ

= (Mgsinθ + μMgcosθ) cosθ

= Mgsinθ cosθ + μMgcos^{2}θ

Net force in vertical direction = Fsinθ

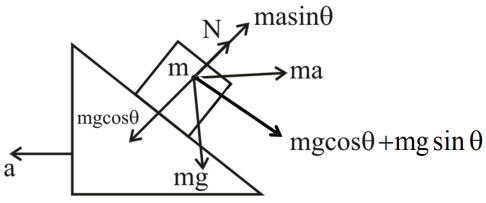
= (Mgsinθ + μMgcosθ) sinθ

= Mgsin^{2}θ + μMgcosθ sinθ

Net force along the plane

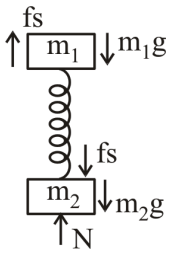
= F = Mgsinθ + μMgcosθ

Net force perpendicular to plane = 0

16) 

N + ma sin θ = mg cos θ

[N = 0] $\left[a = \frac{g}{\tan\theta}\right]$



17) Before

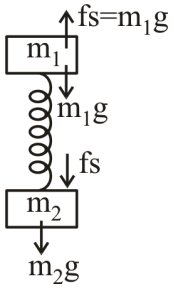
$fs = m_1g$

$N = fs + m_2g$

After table is removed

N = 0

fs do not change



for m_1 $m_1g - m_1g = 0$

$a_{m_1} = 0$

for m_2 $m_2a = m_2g + m_1g$

$$\boxed{a_{m_2} = \frac{(m_1 + m_2)g}{m_2}}$$

18)

Impulse = |change in momentum|

$= \left|m\left(\sqrt{2gh} - 0\right)\right|$

$= \left|2\left(\sqrt{2 \times 10 \times 5} - 0\right)\right|$

= 20 N - S

19)

W = (10) (0.15 sin60°) = $0.75\sqrt{3}$ J

20) $S = \frac{t^3}{3}$

$\Rightarrow$ $dS = t^2dt$

$\Rightarrow a = \frac{d^2S}{dt^2} = \frac{d^2}{dt^2}\left[\frac{t^3}{3}\right]$

= 2t m/s^2

Now work done by the force

$W = \int_0^2 F.dS = \int_0^2 ma.ds$

$\int_0^2 3 \times 2 \times t^2dt \int_0^2 6t^3dt = \frac{3}{2}\left[t^4\right]_0^2 = 24$ J

21)

v = u + at $P = ma$

v = at ...(1) $a = \frac{P}{m}$ ...(2)

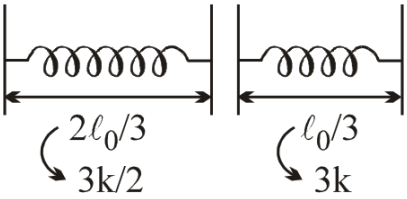
$v = \boxed{\frac{P}{m} \times t}$

KE = $\frac{1}{2}mv^2 = \frac{1}{2} \times m \times \frac{P^2}{m^2} \times t^2$

$\boxed{KE = \frac{P^2t^2}{2m}}$

22)

When a spring is cut into two parts each part has spring constant more than that of original spring. If k = spring constant & ℓ_0 = natural length, then for cut parts



If they are stretched by same amount then work done in shorter part will be double than that in the case of longer part.

Because, $W = \frac{1}{2}kx^2$

23)

$W_f = \Delta$ k.E

$-(4\ mg)\ 30 = \frac{1}{2}m\left(10^2 - 20^2\right)$

$\mu = \frac{1}{2} = 0.5$

24) $\vec{F} = x^2\hat{i} + y^2\hat{j}$ and we need to displace the particle from (0, 0) to (0, 0)
then,

$w = \int \vec{F}.d\vec{r}$ $\quad w = \int_0^0 F_x dx + \int_0^0 F_y dy$

$w = \int_0^0 x^2 dx + \int_0^0 y^2 dy$

$w = 0$

As force is conservative so $W\Delta = 0$
This is because the work done by a conservative force does not depend on path taken.

25) $f(x) = -\frac{dU}{dx}(x)$

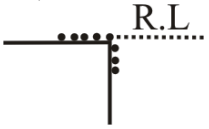
or $U(x) = -\int F(x)\,dx$

Here F(x) = –kx, where k is a positive constant.

U(x) = $+k\int xdx = \frac{1}{2}kx^2$

i.e. parabola

26) Loss in P.E = Gain in K.E



$$U_i - U_f = \frac{1}{2}Mv^2 - 0$$

$$-\left(\frac{M}{3}g\frac{L/3}{2}\right) - \left(-\frac{MgL}{2}\right) = \frac{1}{2}Mv^2$$

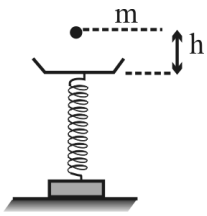
$$\frac{MgL}{2} - \frac{MgL}{18} = \frac{1}{2}Mv^2$$

$$\frac{8}{18}MgL = \frac{1}{2}Mv^2$$

$$\frac{8}{9}MgL = Mv^2$$

$$v = \sqrt{\frac{8}{9}gL}$$

27)



$W_g + W_S = 0$

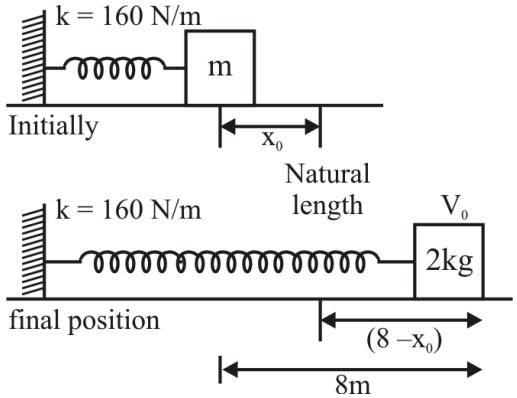
$$mg(h + x) - \frac{1}{2}kx^2 = 0$$

$$K = \frac{2mg(h+x)}{x^2}$$

28)

$$mg \text{ Sin } 30° (d + x) - \frac{1}{2} Kx^2 = 0$$

d = 37.5 cm



29)

From work energy theorem

$$\frac{1}{2}kx_0^2 = \mu mg(8) + \frac{1}{2}k(8 - x_0)^2$$

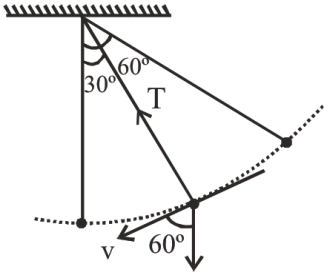
$$\frac{1}{2} \times 160 \times x_0^2 = 0.2 \times 20 \times 8 + \frac{1}{2} \times 160(8 - x_0)^2$$

$80x_0^2 = 32 + 80(8 - x_0)^2$

$80x_0^2 - 80(8 - x_0)^2 = 32$

$[x_0^2 - (8 - x_0)^2] = \frac{2}{5}$

$\Rightarrow \quad 16x_0 = 64.4 \Rightarrow x_0 = 4.025$ m



30)

Power of tension = 0

Power of (mg) = (mg) (v) cos 60°

$v = \sqrt{2gh}$

h = □ (cos30° – cos60°) = 0.71/2

$v = \sqrt{7.1} = 2.6$

Hence power of (mg) = $1 \times 9.8 \times 2.6 \times \frac{1}{2}$

= 13.4 W

31) $X_{cm} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$

$\Rightarrow O = \frac{(2)(1) + (1)(-4) + 5x_3}{2 + 1 + 5}$

$\Rightarrow 5x_1 + 2 - 4 = 0$

$\boxed{x_1 = \frac{2}{5}}$

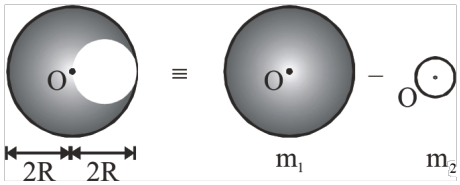
$Y_{cm} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$

$\Rightarrow O = \frac{2(-2) + 1(3) + 5y_3}{2 + 1 + 5}$

$\Rightarrow -4 + 3 + 5y_3 = 0$

$\Rightarrow 5y_3 = 1$

$\boxed{y_3 = \frac{1}{5}}$

32) 

Mass of disc ∝ area

$m_1 \propto \pi (2R)^2$, $m_2^2 \propto R^2$

$m_1 = 4m$, $m_2 = m$

c.m. of the remaining disc

$x_{c.m.} = \dfrac{m_1x_1 - m_2x_2}{m_1 - m_2} = \dfrac{4m \times 0 - m \times R}{4m - m} = -\dfrac{R}{3}$

c.m. of the remaining disc will be at distance R/3, left of O.

33) $y_{cm} = \dfrac{\dfrac{m}{2} \times \dfrac{3R}{8} + \dfrac{m}{2} \times \left(\dfrac{3R}{8} + R\right)}{m}$

$\dfrac{3R}{16} + \dfrac{3R}{16} + \dfrac{R}{2}$

$= \dfrac{3R + 4R}{8} = \dfrac{7R}{8}$

34)

Let m_1 and m_2 are masses and $\vec{v}_1$, $\vec{v}_2$ are velocities with respect to centre of mass. Then

$m_1\vec{v}_1 + m_2\vec{v}_2 = 0$

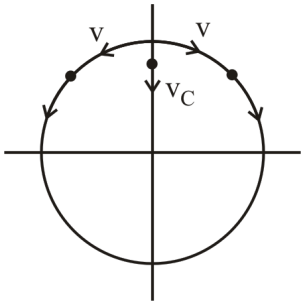
$\vec{v}_2 = -\dfrac{m_1\vec{v}_1}{m_2}$ ...(1)

$v_2 = \dfrac{m_1}{m_2}v_1$ (if $m_1 > m_2$ then)

$v_1 < v_2$ ...(2)

If $v_1 = 0$ then $v_2 = 0$ ...(3)

35)



v_C = component of V along diameter

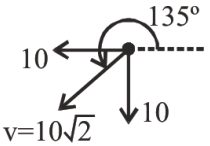
here v_C = V'

V' < V

36) $5m \times 0 = m30\hat{i} + m30\hat{j} + 3m\vec{v}$

$\vec{v} = \dfrac{-30\hat{i} - 30\hat{j}}{3} = -10\hat{i} - 10\hat{j}$

$v = 10\sqrt{2}$ m/s



37) $V_1 = \dfrac{2m - em}{2m + m}V + 0 = \dfrac{V}{3}$

$\Rightarrow e = 1$

38) $\Delta K = \frac{1}{2}\frac{m.2m}{m_1 + m_2}(1-0)v^2 = \frac{mv^2}{3}$

Initial kinetic energy. $K = \frac{1}{2}mv^2$

Percentage loss : $\frac{\Delta K}{K}.100 = \frac{\frac{mv^2}{3}}{\frac{1}{2}mv^2}.100 = \frac{2}{3}.100 = \frac{200}{3}\%$

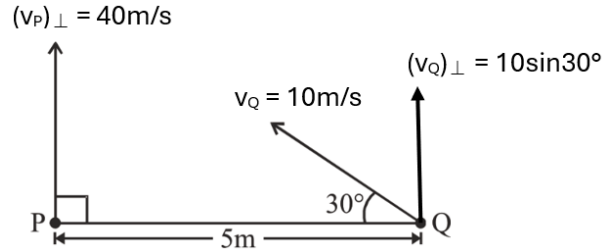
39) Ball (1) : $u_1 = -4$ m/s, $V_1 = ?$
Wall (2) : $u_2 = +20$ m/s, $V_2 = +20$ m/s

$\therefore e = \frac{1}{2} = \frac{20 - V_1}{-4 - 20}$

$V_1 = 32$ m/s

40)

Given diagram



$W_{PQ} = \frac{(v_P)_\perp - (v_Q)_\perp}{\text{distance between 'P' and 'Q'}}$

$= \frac{40 - 10\sin 30^\circ}{5} = \frac{40 - 10 \times \frac{1}{2}}{5}$

$= \frac{40-5}{5} = 7$ rad/sec.

41) $a_T = \frac{dV}{dt} = 4t + 1$

$a_C = \frac{V^2}{r}$

$a = \sqrt{a_c^2 + a_T^2} = 10$ m/s^2

42) $\mu mg = \sqrt{(ma_t)^2 + (ma_c)^2}$

$\mu g = \sqrt{a_t^2 + a_c^2}$ [$\because a_t = \alpha r$]

$25 = \left(\frac{1}{\sqrt{2}}\right)^2 + a_c^2$

$$\frac{49}{2} = a_c^2$$
$$a_c = \frac{7}{\sqrt{2}} = \omega^2(1)$$
$$\frac{7}{\sqrt{2}} = 2\alpha\theta = 2\left(\frac{1}{\sqrt{2}}\right)\theta \Rightarrow \theta = \frac{7}{2}$$
$$2\pi N = \frac{7}{2}$$
$$N = \frac{7}{4\pi}$$

43)

Initial potential energy = mgR

Final kinetic energy = $\frac{1}{2}$ mv^2

Equation: $mgR = \frac{1}{2} mv^2$

$v = \sqrt{2gR}$

44)

$v_b = \sqrt{8gR}$

At top (height difference 2R):

$v_t^2 = v_b^2 - 4gR = 8gR - 4gR = 4gR$

Tension at bottom:

$$T_b - mg = \frac{mv_b^2}{R} \Rightarrow T_b = mg + \frac{m(8gR)}{R} = mg + 8mg = 9mg$$

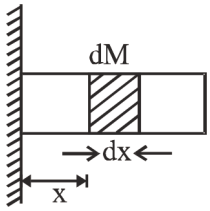
Tension at top :

$$T_t - mg = \frac{mv_t^2}{R} \Rightarrow T_t = \frac{m(4gR)}{R} - mg = 4mg - mg = 3mg$$

Ratio:

$T_t : T_b = 3mg : 9mg = 1:3$

45)



$dM = \left(\frac{M}{L}\right) dx$

force on dM mass is

$dF = (dM)\ \omega^2 x$

$$\Rightarrow F = \int_0^L \frac{M}{L}\omega^2 x dx = \frac{1}{2}M\omega^2 L$$